

## Robotics involves practical skills backed by higher-end theory. It calls for innovative thinking, application and creativity—skills which are not found in textbooks

tems, defines robotics as an amalgamation of the latest engineering breakthroughs in embedded systems, ICT, mechanical product design, bio-engineering, computer science and nanotechnology among others.

### Industry optimistic about 2011

Robots are used in a diverse range of industries today, including automotive, aerospace, electronics and computers, industrial machinery, telecommunications, medicine, manufacturing, agriculture, mining and textiles. More than 120,000 industrial robots were shipped in 2010, with electronics, automotive and metal industries being the main sales drivers. The trend is likely to continue into 2011 and beyond with a further 10-15 per cent increase in robot sales.

The notion that the prospects for the robotics industry are looking up in 2011 is also shared by others in the supply chain, not just robot manufacturers.

According to the IFR (International Federation of Robotics) Quarterly Statistics on Industrial Robots, the robotics industry is capitalising on the increasing demand for automation, especially in the growing Asian markets with China on top. Though robot sales to India are still not much, it is a promising market with great demand for industrial robots in a wide range of industries.

### Opportunities for students

According to Pulkit Gaur, chief technology officer and founder, Gridbots Technologies, "Robotics is going to be one of the biggest industrial catalysts and major factors for growth in production and material handling facilities. Engineering students can avail this opportunity by selecting this as

one of the career options."

With time, there is going to be an increase in the job availability in this sector. "Most of the production, manufacturing and material handling industries are looking for alternatives to increase the productivity, which will soon get reflected in the jobs sector with the adoption of robotic systems in these industries," quips Gaur.

Robotics can be applied to a variety of tasks that we need to perform within and outside our home. "Apart from special applications like handling hazardous situations and tasks that need high accuracy and speed, a new area of robotics called 'service robotics' has begun to develop. It targets medical, space, underwater and housekeeping applications," shares Jayakrishnan T.,

director and country manager, Energid Robotics & Machines Vision.

"Obviously, these applications may demand smart vision, smooth fluid motion, interaction with the environment and artificial intelligence. This must create a very favourable situation for students (who are seeking a career in robotics) to apply their skills in order to support the developments in this connection," he adds.

"India as a country is poised for robotics and automation engineering revolution. Every industry in India is looking for some form of automation technologies to become competitive," says Dr Bijay Kumar Rout, assistant professor, mechanical engineering department and coordinator, Center for Robotics and Intelligent Systems (CRIS), BITS Pilani.

### Know your role

Robotics is multi-disciplinary and so are the job opportunities in this area. This integrative field offers jobs in

three major sectors—design and manufacturing of robotic systems, embedded and control system, and software control and simulation environments for these robotic systems. A lot of responsibilities do come up while working in these sectors. As compared to traditional systems where machine components are decoupled, in this field all the systems function in tandem with other components. So while designing or working on robotic systems, one has to take care of a lot of factors, cautions Gaur.

Robotics engineering combines elements of mechanical, electrical and software engineering in the design, testing and development



Automatic fitting of windscreen by robots